

Week 4 – Data Visualization & Complete Data Analysis Project

E-commerce Sales Analysis

1. Project Overview

This project was completed as part of Week 4 of my Data Science Internship at The Developers Arena.

The objective of this project is to perform a complete data analysis workflow using a real sales dataset. The project covers data loading, data inspection, data cleaning, exploratory analysis, sales analysis, data visualization, and generation of meaningful business insights.

Python, Pandas, and Matplotlib were used to analyze the dataset and create professional visualizations.

The project demonstrates the complete data analysis pipeline:

****Load → Clean → Analyze → Visualize → Interpret****

2. Project Objectives

The main objectives of this project are:

- Load and work with a real-world sales dataset.
- Explore the structure and contents of the dataset.
- Inspect rows, columns, and data types.
- Check the dataset for missing values and duplicate records.
- Clean and validate the data where required.
- Calculate important sales metrics.
- Analyze sales performance by product.
- Analyze sales performance by region.
- Analyze sales trends over time.
- Create meaningful data visualizations.
- Generate written business insights from the analysis.
- Handle common errors and invalid inputs appropriately.
- Document the complete project professionally.

3. Project Topic

E-commerce Sales Analysis

The project analyzes sales transaction data to understand:

- Product performance
- Sales revenue
- Quantity sold

- Regional performance
- Sales distribution
- Sales trends over time

The analysis is based on the actual sales dataset provided for the internship.

4. Dataset Description

The project uses the sales dataset provided for the internship.

The dataset contains sales transaction records with information about products, quantities, prices, customers, regions, dates, and total sales.

Dataset Columns

Column	Description
Date	Date on which the transaction occurred
Product	Name/category of the product sold
Quantity	Number of units sold
Price	Price associated with the product
Customer_ID	Identifier of the customer
Region	Region where the sale occurred
Total_Sales	Total sales/revenue generated by the transaction

The dataset is stored inside the project's `data` directory.

5. Technologies Used

The following technologies and tools were used:

- Python 3.x
- Pandas
- Matplotlib
- Visual Studio Code
- Git
- GitHub

Python Libraries

Pandas

Pandas was used for:

- Reading the CSV file
- Data inspection
- Data cleaning

- Grouping and aggregation
- Statistical calculations
- Sales analysis

Matplotlib

Matplotlib was used to create the data visualizations.

6. Project Structure

The project follows a clean and organized folder structure:

```
```text
```

```
Task_4_DataVis_1stProject/
```

```
|
|— main.py
|— requirements.txt
|
|— data/
| └─ sales_data.csv
|
|— visualizations/
| └─ sales_by_product.png
| └─ sales_by_region.png
| └─ sales_distribution_by_product.png
| └─ sales_trend.png
|
|— report/
| └─ analysis_report.md
|
└─ screenshots/
```

5. Visualizations

File Edit Selection View Go Run

Week1\_Task

Update

Explorer

main.py Task\_4\_DataVis\_1st... U

sales\_data.csv Task\_4\_Data... U

Week1\_Task

Task\_2\_Student\_Grade\_Calculator

Task\_3\_Introduction\_to\_DataAnal...

Task\_4\_DataVis\_1stProject

\_\_pycache\_\_

data

report

screenshots

visualizations

sales\_by\_product.png U

sales\_by\_region.png U

sales\_distribution\_by\_p... U

sales\_trend.png U

main.py U

requirements.txt U

Outline

Timeline

BMD Command

Application Builder

CS-Script - Active

Servers

main.py U sales\_data.csv U

Task\_4\_DataVis\_1stProject > data > sales\_data.csv > data

1 Date,Product,Quantity,Price,Customer\_ID,Region,Total\_Sales

2 2024-01-01,Phone,7,37300,CUST001,East,261100

3 2024-01-02,Headphones,4,15406,CUST002,North,61624

4 2024-01-03,Phone,2,21746,CUST003,West,43492

5 2024-01-04,Headphones,1,30895,CUST004,East,30895

6 2024-01-05,Laptop,8,39835,CUST005,North,318680

Problems

Debug Console

Ports

Spell Checker 36

Code Reference Log

Terminal

D:\Developer\_Arena\Week1\_Task\Task\_4\_DataVis\_1stProject>python main.py

First 5 rows:

0 Date Product Quantity Price Customer\_ID Region Total\_Sales

0 2024-01-01 Phone 7 37300 CUST001 East 261100

1 2024-01-02 Headphones 4 15406 CUST002 North 61624

2 2024-01-03 Phone 2 21746 CUST003 West 43492

3 2024-01-04 Headphones 1 30895 CUST004 East 30895

4 2024-01-05 Laptop 8 39835 CUST005 North 318680

Dataset shape: (100, 7)

Columns: ['Date', 'Product', 'Quantity', 'Price', 'Customer\_ID', 'Region', 'Total\_Sales']

Data types:

Date datetime64[ns]

Product object

Quantity int64

Price int64

File Edit Selection View Go Run

Week1\_Task

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6 2024-01-05,Laptop,8,39835,CUST005,North,318680

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Terminal

Data types:

Date datetime64[ns]

Product object

Quantity int64

Price int64

Customer\_ID object

Region object

Total\_Sales int64

dtype: object

Basic statistics:

count Date Quantity Price Total\_Sales

count 100 100.000000 100.000000 100.000000

mean 2024-02-19 12:00:00 4.780000 25808.510000 123650.480000

min 2024-01-01 00:00:00 1.000000 1308.000000 6540.000000

25% 2024-01-25 18:00:00 2.750000 14965.250000 39517.500000

50% 2024-02-19 12:00:00 5.000000 24192.000000 97955.500000

main\*

Launchpad

0 0

Live Share

AWS

Amazon Q

BLACKBOX Agent

CSVlint

Query

Align

Open Website

Quokka

Prettier

File Edit Selection View Go Run ... Week1\_Task Update

Explorer

- Open Editors
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```

Problems Debug Console Ports Spell Checker 36 Code Reference Log

Terminal

```
count Date Quantity Price Total_Sales
mean 2024-02-19 12:00:00 4.780000 25808.510000 123650.400000
min 2024-01-01 00:00:00 1.000000 1308.000000 6540.000000
25% 2024-01-25 18:00:00 2.750000 14965.250000 39517.500000
50% 2024-02-19 12:00:00 5.000000 24192.000000 97955.500000
75% 2024-03-15 06:00:00 7.000000 38682.250000 175792.500000
max 2024-04-09 00:00:00 9.000000 49930.000000 373932.000000
std 2.588163 13917.630242 100161.085275
```

=====

WEEK 4 - COMPLETE SALES ANALYSIS PROJECT

=====

Rows: 100 | Columns: 7

Missing values found: 0

Duplicate rows found: 0

Total sales: 12,365,048.00

File Edit Selection View Go Run ... Week1\_Task Update

Explorer

- Open Editors
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```

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Terminal

```
=====
```

WEEK 4 - COMPLETE SALES ANALYSIS PROJECT

=====

Rows: 100 | Columns: 7

Missing values found: 0

Duplicate rows found: 0

Total sales: 12,365,048.00

Average sale: 123,650.48

Maximum sale: 373,932.00

Minimum sale: 6,540.00

Total quantity sold: 478

Best-selling product: Laptop

Sales by product:

Product	Sales
Laptop	3889210
Tablet	2884340

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```

Problems Debug Console Ports Spell Checker 36 Code Reference Log

Terminal

```
Phone 2859394
Headphones 1384033
Monitor 1348071
```

Sales by region:

Region	Sales
North	3983635
South	3737852
East	2519639
West	2123922

Charts saved:

- visualizations\sales\_by\_product.png
- visualizations\sales\_trend.png
- visualizations\sales\_by\_region.png
- visualizations\sales\_distribution\_by\_product.png